Connor Zhizhen Lin

Education

2020–2024 **PhD in Computer Science**, *Stanford University*, Stanford, CA.
Advisors: Leonidas Guibas, Gordon Wetzstein
Thesis: 3D Neural Representations for Controllable Editing and Generation
Stanford Graduate Fellow (*David Cheriton*)

2018–2019 **MSc in Computer Science**, *Carnegie Mellon University*, Pittsburgh, PA.
Advisor: Keenan Crane
Thesis: Periodic Conformal Parameterization

2015–2018 **BSc in Computer Science**, *Carnegie Mellon University*, Pittsburgh, PA.

Experience

2024-2026 **Research Lead**, *Luma AI*, Redwood City, CA.
- Leading pre-training and post-training for Luma's `OPAL` effort to train large-scale, generalizable world action models for robotics (SOTA results to be released early Nov. 2026)
- Led pre-training and post-training for Luma's `UNI-1` multimodal image model. Led follow-up research efforts for UNI storyboard (`Luma Scenes`), manga, and super-resolution.
- Led post-training for Luma's `Photon` image model and `Ray2` video model.

2023-2024 **Co-founder & CTO**, *Apparate AI [Acquired by Luma AI]*, Palo Alto, CA.
- Co-founded Apparate to develop multimodal generative AI capabilities via controllable visual embodiments of LLMs and conversational entities.
- Developed and released PROTEUS, a real-time video foundation model for generating highly realistic and expressive humans.

2023 **PhD Student Researcher**, *Google Research*, Mountain View, CA.
- Researched new techniques for representing 3D humans and objects with dynamic 3D gaussians on the 3D Scene Understanding team within Google Perception.

2022-2023 **PhD Research Intern**, *NVIDIA Research*, Toronto.
- Developed a new method for single-shot 3D reconstruction and animation of neural avatars (SIGGRAPH 2023) that combines implicit SDF representations with explicit UV-parameterized texture maps.

2021 **PhD Research Intern**, *Adobe Research*, London.
- Developed NeuForm (NeurIPS Oral 2022), a novel hybrid approach combining overfitting and generalizable model priors for neural 3D editing.

2019-2020 **Software Engineer**, *Google*, Mountain View, CA.
- Researched and prototyped end-to-end solutions for real-time depth inference and improved performance of depth inference in Portrait mode.

2018 **Software Engineering Intern**, *Google Daydream*, New York, NY.
- Implemented a VR Unity plugin using C# and C++ that dynamically recognizes and morphs user virtual handwriting into VR text.

2017 **Software Engineering Intern**, *Yahoo!*, Sunnyvale, CA.

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Teaching

Teaching Experience

- Teaching Assistant (Fall 2024). Artificial Intelligence: Principles and Techniques (CS221)
- Teaching Assistant (Fall 2023). Computer Graphics: Animation and Simulation (CS248B)
- Teaching Assistant (Fall 2017, Fall 2018, Spring 2019). Computer Graphics (15-462/15-662)
- Teaching Assistant (Spring 2017). Principles of Imperative Computation (15-122)

Publications

SAP 2024 **GazeFusion: Saliency-Guided Image Generation.**
*Y. Zhang, N. Wu, **C. Z. Lin**, G. Wetzstein, Q. Sun*

SIGGRAPH 2023 **Single-Shot Implicit Morphable Faces with Consistent Texture Parameterization.**
***C. Z. Lin**, K. Nagano, J. Kautz, U. Iqbal, L. Guibas, G. Wetzstein, S. Khamis*

NeurIPS 2022 (Oral) **NeuForm: Adaptive Overfitting for Neural Shape Editing.**
***C. Z. Lin**, N. J. Mitra, G. Wetzstein, L. Guibas, P. Guerrero*

ECCVW 2022 (Learn3DG) **3D GAN Inversion for Controllable Portrait Image Animation.**
C. Z. Lin, D. B. Lindell*, E. R. Chan, G. Wetzstein*

CVPR 2022 (Oral) **EG3D: Efficient Geometry-aware 3D Generative Adversarial Networks.**
E. R. Chan, **C. Z. Lin***, M. A. Chan*, K. Nagano*, B. Pan, S. D. Mello, O. Gallo, L. Guibas, J. Tremblay, S. Khamis, T. Karras, G. Wetzstein*

SIGGRAPH 2021 **ACORN: Adaptive Coordinate Networks for Neural Representation.**
J. N. P. Martel, D. B. Lindell*, **C. Z. Lin**, E. R. Chan, M. Monteiro, G. Wetzstein*

Masters Thesis **Periodic Conformal Parameterization.** *SCS Technical Report*
C. Z. Lin

Talks

Aug. 2023 **Single Shot Implicit Morphable Faces** *SIGGRAPH* Conference Talk

Mar. 2023 **Neural Avatars from a Single Image** *Eyeline Studios* Invited Research Talk

Nov. 2022 **Local Neural Shape Editing** *NeurIPS* Conference Talk

Jun. 2022 **Advancing and Applying 3D GANs** *Stanford University* CS PhD Qualifying Exam

Jul. 2019 **Periodic Conformal Parameterization** *Carnegie Mellon University* Masters Thesis Defense

Dec. 2017 **Real World Fabrication of 3D Meshes** *Carnegie Mellon University* SCS Undergraduate Research Showcase

Awards

- Stanford Graduate Fellowship (David Cheriton)
- 5x Dean's List
- CMU University Honors

Service

2022-2023 **Reviewer** *CVPR, SIGGRAPH, SIGGRAPH Asia*

2022 **Stanford Club Badminton** *President*

2021 **Stanford CS PhD Admissions** *Committee Member*

2021 **Stanford Club Badminton** *Financial Officer*

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